



UNIVERSITÀ DI PARMA

Problem Solving

*If I only had an hour to chop down a tree, I
would spend the first 45 minutes sharpening
my axe.*

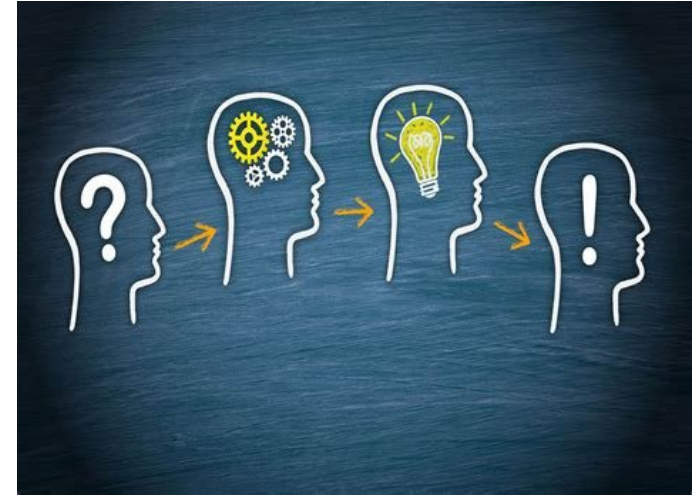
Abraham Lincoln

- What is problem solving?
- Why it is important?
- Few steps
 - Understand the problem
 - Sample data
 - Pseudocode
 - Code
 - Debug or Optimization

SUMMARY

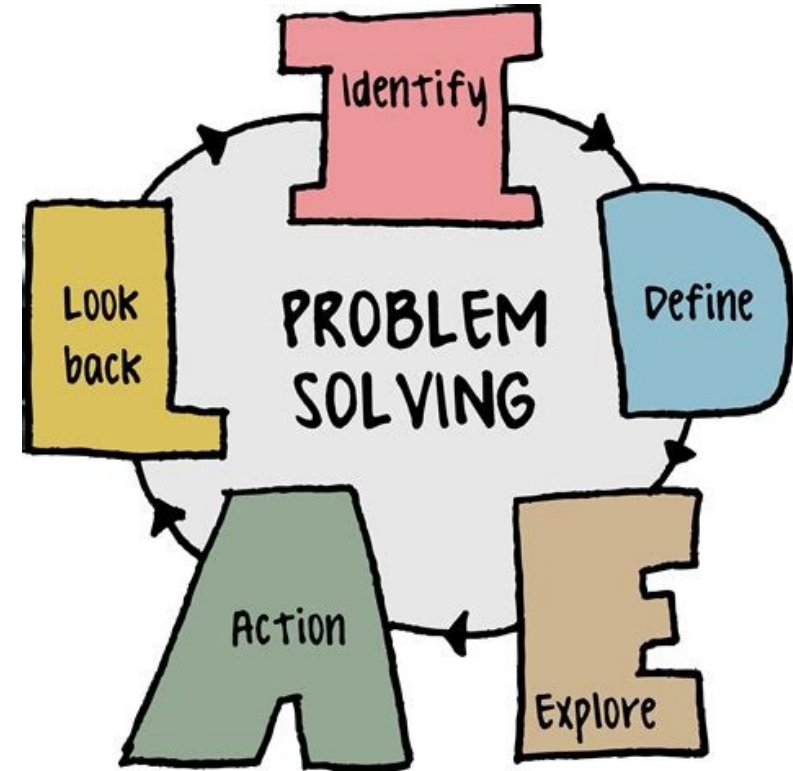


- What is?
 - Transform the description of a problem into a solution!
- Why it is important?
 - Engineers use problem solving skills, **constantly**
 - Not computer programming only!
 - Direct impact on the pace of your professional progress and career growth



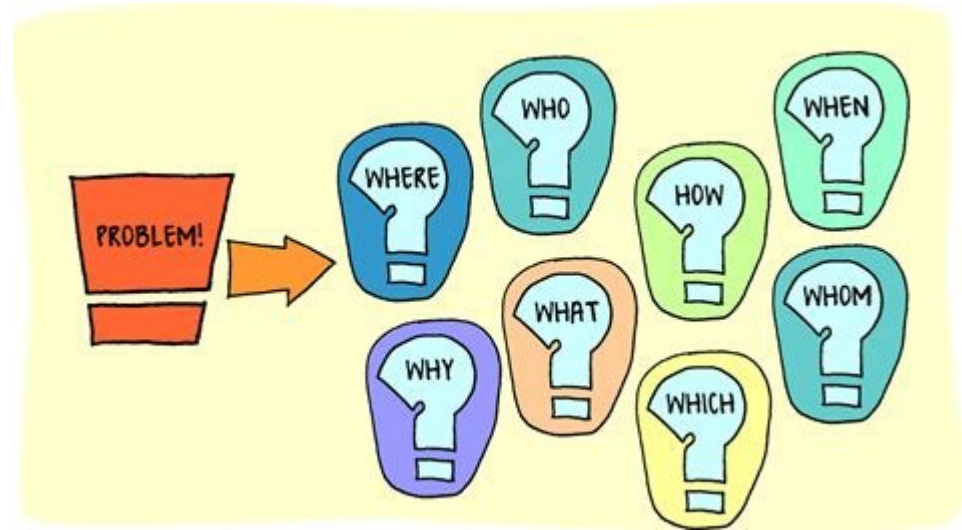
What is the right approach?

- Do not rush in coding!
 - If you have a time limit this can appear captivating
 - But it is wrong!
- 4 steps to follow in the next slides



1. Understand the problem: read

- Again, do not rush in coding!
- You can not solve a problem you do not understand
- Read through the problem
- Clarify any part
 - Maybe ask or discuss!
- Once you have read
 - Read it again
 - Up to three times...



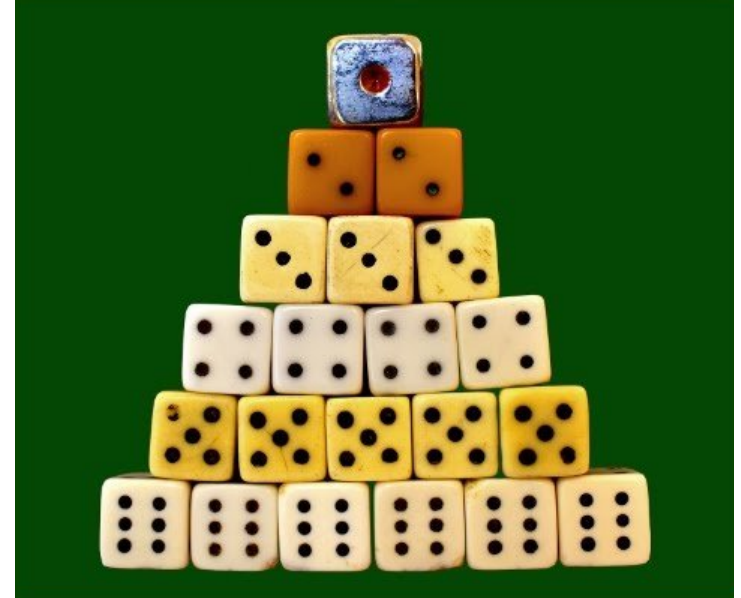
1. Understand the problem: simulate

- Work through the problem manually
 - What are the inputs?
 - What are the outputs?
- Consider sample inputs
- Try to analyze the output
 - Initially simple inputs
 - Then add complexity
 - Also consider corner/edge cases



2. Devise a plan for solving the problem: top-down approach

- Step a step plan to solve the problem
- Think about how you would solve it as a human
- Break down the problem
 - Small sub-problems or chunks
 - Solve them one by one
 - Connect the solution of each sub problem as input for others
- Edge and corner problems are dealt by the proposed solution?



2. Devise a plan for solving the problem: pseudocode

- Initially write down your steps in natural language
- Then translate your steps in pseudocode
 - Having final language in mind can help

```
MOT1
Program to generate PWM output to Motor

  Initialize
    Outputs
      Motor
    Inputs
      Speed up
      Speed down
      Run enable
    Registers
      Count = 128

  Start loop

    IF Run enable = off THEN wait
    IF Speed up = on THEN inc Count
    IF Count = 0 THEN dec Count
    IF Speed down = on THEN dec Count
    IF Count = 0 THEN inc Count

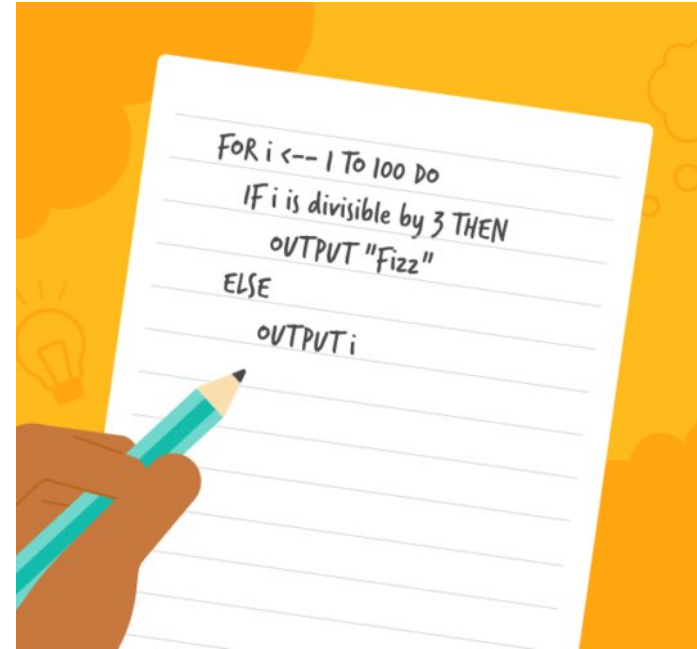
    Switch on Motor
    Delay for Count

    Switch off Motor
    Delay for 256-Count

  End loop
```

3. Carry out the plan: write actual code

- Translate the pseudocode or algorithm description in programming language
 - Eventually you can code!
- Do not optimize code too early
 - Not required for us...



3. Carry out the plan: debug

- For sure the initial release will not work
- Debug
 - Look at compiler warnings
 - Add some print
 - Again, independently debug each part



4. Look back over what you've done: refactoring

- Not necessary in our case
- Reflect on your solution and, maybe, simplify your steps
 - Are improvements possible?
 - What other approaches are possible?
 - Is there anything you can generalize?





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**KEEP
CALM
IT'S
QUESTION
TIME**

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